MASOOD NAZARI

Senior Data Engineer | ETL Pipeline Automation | Healthcare & FinTech

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LinkedIn GitHub Portfolio

PERSONAL PROFILE

Senior Data Engineer (MSc Distinction) with 5+ years of experience designing and deploying **production-grade ETL pipelines**, **data quality frameworks**, and **API-driven data architectures** across regulated healthcare and FinTech environments. Technical lead for pipeline automation at the University of Southampton's Clinical Trials Unit, where engineered solutions reduced data ingestion errors to <**0.1%** and cut manual data processing workloads by **70%+**. Proven expertise in Python, SQL, FastAPI, Docker, and Azure; experienced in building GDPR/GxP-compliant data lineage and audit systems directly transferable to FCA/Basel III and other regulated sectors.

TECHNICAL SKILLS

- **Data Engineering:** ETL/ELT Pipelines, Data Warehouse Design, Schema Mapping & Validation, Data Lineage, Data Quality Frameworks, Automated QA, Data Reconciliation, Pydantic Schema Enforcement.
- **Languages:** Python (Pandas, GeoPandas, Shapely, FastAPI), SQL (PostgreSQL/T-SQL), R, Bash, SAS.
- **Cloud & Infrastructure:** Azure (ACI, GCP), Docker, Terraform (IaC), Google Earth Engine, API Integration.
- **ML & Analytics:** Scikit-learn, XGBoost, CatBoost, Prophet, SHAP/LIME, Neural Networks (BERT, SpaCy).
- **BI & Visualisation:** Streamlit, Power BI, Tableau, R Shiny, Plotly, Excel (Advanced).
- **Compliance:** GDPR, GxP (ICH/FDA), Regulatory Audit Trail Design, Metadata Governance, FCA/Basel III (Transferable).

PROFESSIONAL EXPERIENCE

Data Engineer / Clinical Data Coordinator (Technical Lead) — Sep 2024 – Present

Southampton Clinical Trials Unit (SCTU) | University of Southampton — *Southampton, UK*

Technical lead for pipeline automation, data quality, and reporting infrastructure across the Cancer Vaccine Launch Pad (CVLP) oncology programme — spanning BNT122-01, BNT113-01, and SCOPE trials.

- **ETL Pipeline Architecture (Python/FastAPI):** Designed and deployed a RESTful event-driven data pipeline to automate Adverse Event (AE) reconciliation across 70+ external clinical sites. Implemented **Pydantic schema validation** and error-logging that reduced data ingestion errors to near-zero (<**0.1%**) and flagged anomalies for immediate triage.
- **Real-Time Data Platform:** Engineered a containerised **ETL + reporting layer** (Streamlit/Docker) processing trial data from weekly batch to near-real-time. Enabled Trial Managers to identify data discrepancies **6× faster**, reducing downstream QA correction cycles.
- **Data Lineage & Regulatory Compliance:** Implemented immutable audit-log pipelines satisfying **GDPR/GxP** requirements. Created a data lineage framework that reduced QA retrieval time by **80%** and is directly transferable to FCA/Basel III regulatory audit models.
- **Infrastructure & Security Audit:** Led architectural review of third-party integration tools across the data estate. Identified OAuth authentication vulnerabilities and authored a remediation roadmap securing data access for all external site connections.
- **SAS & R Automation:** Automated statistical report generation for BNT113-01 and SCOPE studies, significantly reducing manual analyst time per reporting cycle.

Data Engineer (Internship) — Jun 2024 – Sep 2024

SC Solutions | Environmental Analytics — *Remote/Hybrid*

- **Spatial ETL Pipeline:** Architected automated **geospatial ETL pipelines** using Python (GeoPandas, Shapely) and Google Earth Engine APIs to ingest, transform, and validate multi-terabyte satellite imagery datasets (NASA GEDI, Sentinel SAR/Optical).
- **Data Quality & Geometry Validation:** Implemented automated **topology and geometry validation scripts** to detect and resolve data errors (gaps, anomalies, invalid geometries) in noisy satellite data before downstream processing.
- **Performance Engineering:** Optimised data preprocessing latency by **80%** through parallelised feature extraction, reducing the full validation-to-model cycle from days to hours.
- **Scalable Deployment:** Scaled spatial ML models across **10,000+ hectares**, ensuring high-fidelity data mapping and integrated Explainable AI (SHAP) for audit-ready carbon stock estimation outputs.

Senior Data Analyst / Data Engineer — Oct 2019 – Sep 2023

Parsian E-Commerce Co. | FinTech — *Tehran*

- **Fraud Detection Pipeline:** Engineered ML-based transaction anomaly detection pipelines improving fraud capture rates by **26%** and reducing financial liability exposure.
- **Reconciliation Automation:** Designed **SQL and Python-based data reconciliation pipelines** to replace manual financial ledger processes. Reduced manual handling time by **43%** and cut the month-end data closing cycle by **2 days**.
- **Data Warehouse & BI:** Built executive-level KPI dashboards on top of structured data warehouse queries, providing real-time visibility into cost drivers and increasing decision-making efficiency by **∼27%**.

Data Analyst / Reporting Automation — Jun 2018 – Sep 2019

MaximTaxi | Private Transport — *Tehran*

- **SQL Data Validation:** Queried large transactional datasets (PostgreSQL) to identify financial discrepancies, improving supplier payment data accuracy by **23%**.
- **Automated Reporting:** Developed automated SQL queries for compliance and tax reporting, eliminating manual Excel-based processes and reducing audit risk.

TECHNICAL ENGINEERING & ARCHITECTURE

Fraud Detection Pipeline API | *Open Source* — 2024

- **Stack:** Python, FastAPI (Async), NetworkX, Docker, Pydantic, Pandas.
- **Pipeline Architecture:** Engineered a **high-concurrency Async FastAPI backend** processing **5 million+ transactions** with sub-100ms latency. Implemented **Pydantic schema validation** for strict type-safe financial data ingestion.
- **Data Quality & Compliance:** Deployed graph theory algorithms (NetworkX) to detect circular fraud patterns. Integrated Explainable AI (XAI) for regulatory-compliant transaction rejection transparency.

BioVision Analytics Hub (Geospatial ETL) | *Open Source* — 2024

- **Stack:** Python, GeoPandas, LightGBM, XGBoost, Sentinel-1/2 APIs, Streamlit, Mapbox.
- **Spatial Data Pipeline:** Architected an **automated ETL pipeline** for multi-terabyte geospatial datasets with parallelised feature extraction and geometry validation. Achieved **0.89 R^2** accuracy for forest ecosystem monitoring.
- **Research Outcome:** Formed the basis of an MSc Distinction dissertation on AI-driven geospatial forecasting at the University of Southampton.

NIHR Research Intelligence Dashboard (Data Quality) | *Open Source* — 2024

- **Stack:** Python, Streamlit, Scikit-learn, Pandas, Plotly.
- **Data Quality Engine:** Built a **Fuzzy Matching-based deduplication pipeline** detecting duplicate funding awards (5.6% rate) and enforcing NIHR data governance rules. Increased dataset completeness score to **73.6%**.
- **Impact:** Automated parliamentary reporting workflows across a **£171.6M** research portfolio, delivering verified **122:1 ROI** in annual cost savings.

InsurePrice: Actuarial Risk Engine (Cloud Architecture) | *Open Source* — 2024

- **Stack:** Python, CatBoost, SHAP, Terraform (IaC), Azure Container Instances (ACI), FastAPI.
- **Cloud Infrastructure:** Deployed a containerised, scalable **Azure** pricing engine using **Terraform for Infrastructure-as-Code**. Implemented full dev/prod parity via Docker with automated type-safe testing.

Bank Marketing Analytics Platform | *Open Source* — 2024

- **Stack:** Python, React, Flask (REST API), Streamlit, Scikit-learn (SMOTE).
- **Architecture:** Designed a dual-interface deployment: Streamlit for rapid prototyping and a decoupled **Flask/React REST API** serving real-time predictions via 7 endpoints. Applied SMOTE for class imbalance, boosting minority class recall to **60%+**.

EDUCATION

MSc Data & Decision Analytics — Distinction — Sep 2023 – Sep 2024
University of Southampton, UK

- **Dissertation:** "AI-driven geospatial forecasting" — Distinction. Deployed as the BioVision Analytics Hub.
- **Core modules:** Advanced Machine Learning, Data Mining, Stochastic Processes, Operational Research.

BSc Materials Engineering — Distinction, Top 5% of cohort — Sep 2010 – Sep 2014
Semnan University

- Specialised in computational modelling and simulation.